Variants of PINN

56262

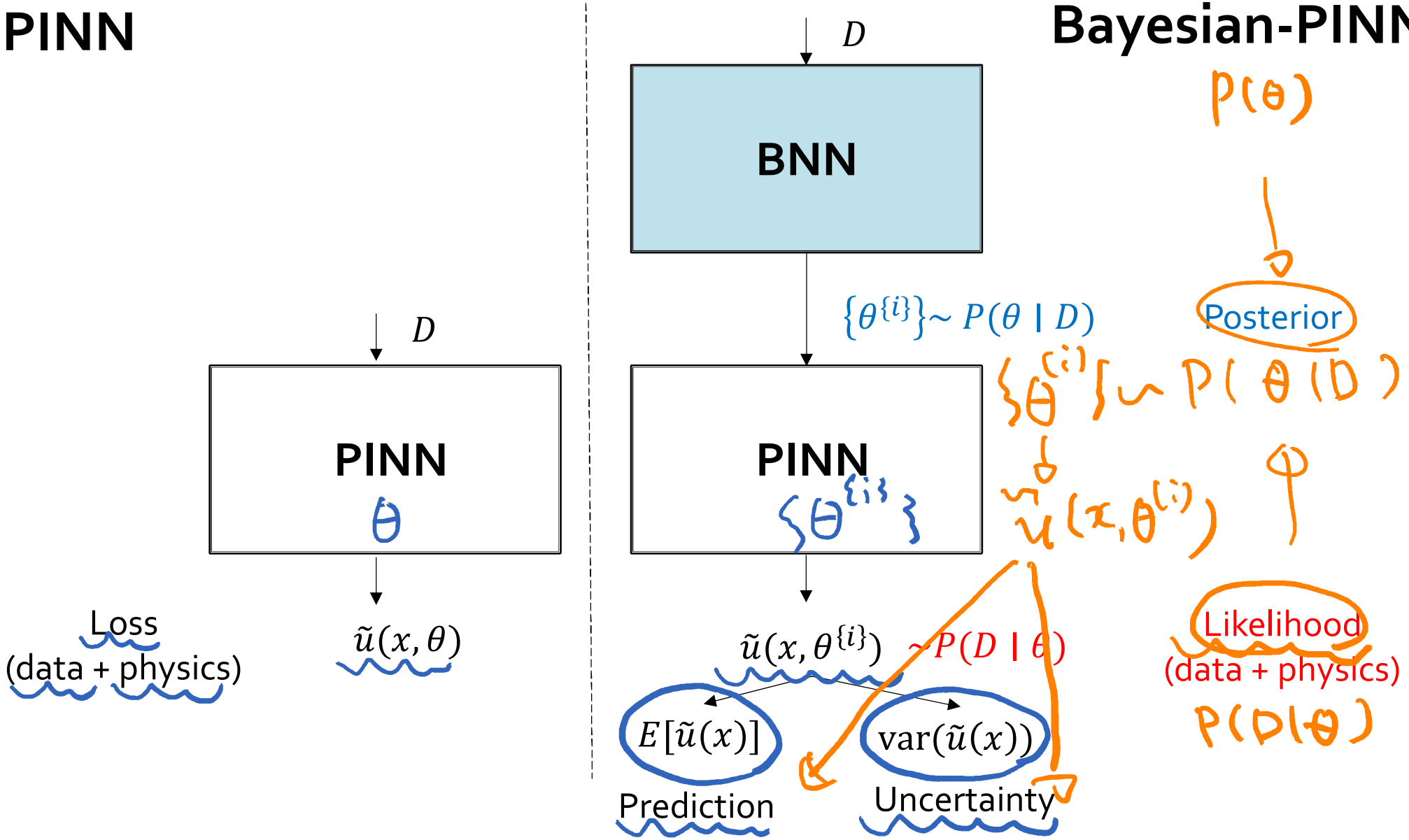
Physics-informed neural networks 2

B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data
(*Journal of Computational Physics, 2021*)

Problem of Vanilla PINN

Problem of Vanilla PINN	Alternatives	Keywords
Instance-wise retraining	Neural Operator	Operator learning
Noisy environments	**Bayesian PINN**	**Uncertainty**
Loss/gradient imbalance	Self-adaptive PINN	Self-adaptive weighting
Unstable training	X-PINN	Domain decomposition

PINN vs. Bayesian PINN 1/2



PINN vs. Bayesian PINN 2/2

	Vanilla PINN	Bayesian-PINN
Output	Deterministic	Probabilistic
Uncertainty	Not available	Available
Noise	Weak	Strong
Overfitting	Possible	Reduced

- **Bayesian-PINN**
  - Bayesian-PINN is the most likely solution with a certain level of uncertainty

Problem Statement

- **Bayesian-PINN :** Uncertainty-aware PDE solver

  = Physical laws (PINN) + Uncertainty (BNN)

  - **Disadvantages of Vanilla PINNs :** Sensitivity to noisy data, Lack of uncertainty quantification, Risk of overfitting
  - **BNN** : quantify the uncertainty arising from the noisy data in the Bayesian framework.
  - **Bayesian-PINNs** : obtain more accurate predictions in scenarios with large noise due to their capability of avoiding overfitting.

- **Core Idea**
  - BNNs : modeling uncertainty in parameters
  - PINNs : enforcing physical consistency

General PDE System

- B-PINNs consider a general PDE system defined

PDE constraint

Boundary condition

$$\mathcal{N}_x(u;\lambda) = f, \quad x \in D$$
$$\mathcal{B}_x(u;\lambda) = b, \quad x \in \partial D$$

- **PDE constraint**: This encodes the physical law governing the system.
- **Boundary condition** : This enforces constraints at the domain boundary.
- The solution is approximated by a neural network: $u(x) \approx \tilde{u}(x;\theta)$

$\mathcal{N}_x$	Differential operator	$\mathcal{B}_x$	Boundary operator
$u(x)$	Unknown solution	$b(x)$	Boundary condition
λ	PDE parameters	D	Domain
$f(x)$	Forcing/source term	∂D	Boundary of the domain

Observation
(u, f and b)

Noisy Environments

- **Noisy Observation** (u, f and b)
  - We consider the scenario where our available dataset D are scattered noisy measurements of u, f and b from sensors.
  - We assume that the measurements are independently Gaussian distributed centered at the hidden real value

$$\bar{u}^{(i)} = u(\boldsymbol{x}_u^{(i)}) + \epsilon_u^{(i)}, \quad i = 1, 2 \ldots N_u,$$
$$\bar{f}^{(i)} = f(\boldsymbol{x}_f^{(i)}) + \epsilon_f^{(i)}, \quad i = 1, 2 \ldots N_f,$$
$$\bar{b}^{(i)} = b(\boldsymbol{x}_b^{(i)}) + \epsilon_b^{(i)}, \quad i = 1, 2 \ldots N_b,$$

noisy data = clean data + error

Bayesian Formulation

- **Likelihood (Physics + Data)**
  - The likelihood combines: data mismatch, PDE residual, boundary conditions

$$P(\mathcal{D}|\boldsymbol{\theta}) = P(\mathcal{D}_u|\boldsymbol{\theta})\, P(\mathcal{D}_f|\boldsymbol{\theta})\, P(\mathcal{D}_b|\boldsymbol{\theta}),$$

$$P(\mathcal{D}_u|\boldsymbol{\theta}) = \prod_{i=1}^{N_u} \frac{1}{\sqrt{2\pi\sigma_u^{(i)2}}} \exp\left(-\frac{(\tilde{u}(\boldsymbol{x}_u^{(i)};\boldsymbol{\theta}) - \bar{u}^{(i)})^2}{2\sigma_u^{(i)2}}\right), \quad \text{Data mismatch}$$

$$P(\mathcal{D}_f|\boldsymbol{\theta}) = \prod_{i=1}^{N_f} \frac{1}{\sqrt{2\pi\sigma_f^{(i)2}}} \exp\left(-\frac{(\tilde{f}(\boldsymbol{x}_f^{(i)};\boldsymbol{\theta}) - \bar{f}^{(i)})^2}{2\sigma_f^{(i)2}}\right), \quad \text{PDE residual}$$

$$P(\mathcal{D}_b|\boldsymbol{\theta}) = \prod_{i=1}^{N_b} \frac{1}{\sqrt{2\pi\sigma_b^{(i)2}}} \exp\left(-\frac{(\tilde{b}(\boldsymbol{x}_b^{(i)};\boldsymbol{\theta}) - \bar{b}^{(i)})^2}{2\sigma_b^{(i)2}}\right). \quad \text{Boundary conditions}$$

Bayesian Formulation

- **Prior (BNN)**
  - The neural network parameters θ are treated as random variables: $P(\theta)$
- **Posterior**
  - Using Bayes' theorem:

$$P(\theta|D) \propto P(D|\theta)P(\theta)$$

posterior ∝ likelihood × prior

- **Posterior Sampling**
  - $\{\theta^{\{i\}}\} \sim P(\theta \mid D)$
  - Multiple prediction (probability distribution of $\tilde{u}$): $\tilde{u}(x, \theta^{\{i\}})$
  - Mean $E[\tilde{u}(x)]$ → **Prediction**, Variance $\mathrm{var}(\tilde{u}(x))$ → **Uncertainty**

Pipeline 1/2

[Input : Noisy data (u, f, b)]

D_u: solution data, D_f: forcing data, D_b: boundary data

[Neural Network]

$\tilde{u}(x, \theta)$

[Bayesian Formulation]

Prior $P(\theta)$: Gaussian prior

Likelihood $P(D|\theta)$: Exponential function

$P(\theta|D) \propto P(D|\theta) \times P(\theta)$: Bayes' Theorem

[Sampling]

HMC / VI : $\{\theta^{\{i\}}\} \sim P(\theta \mid D)$

$\theta_1 \rightarrow u_1(x)$

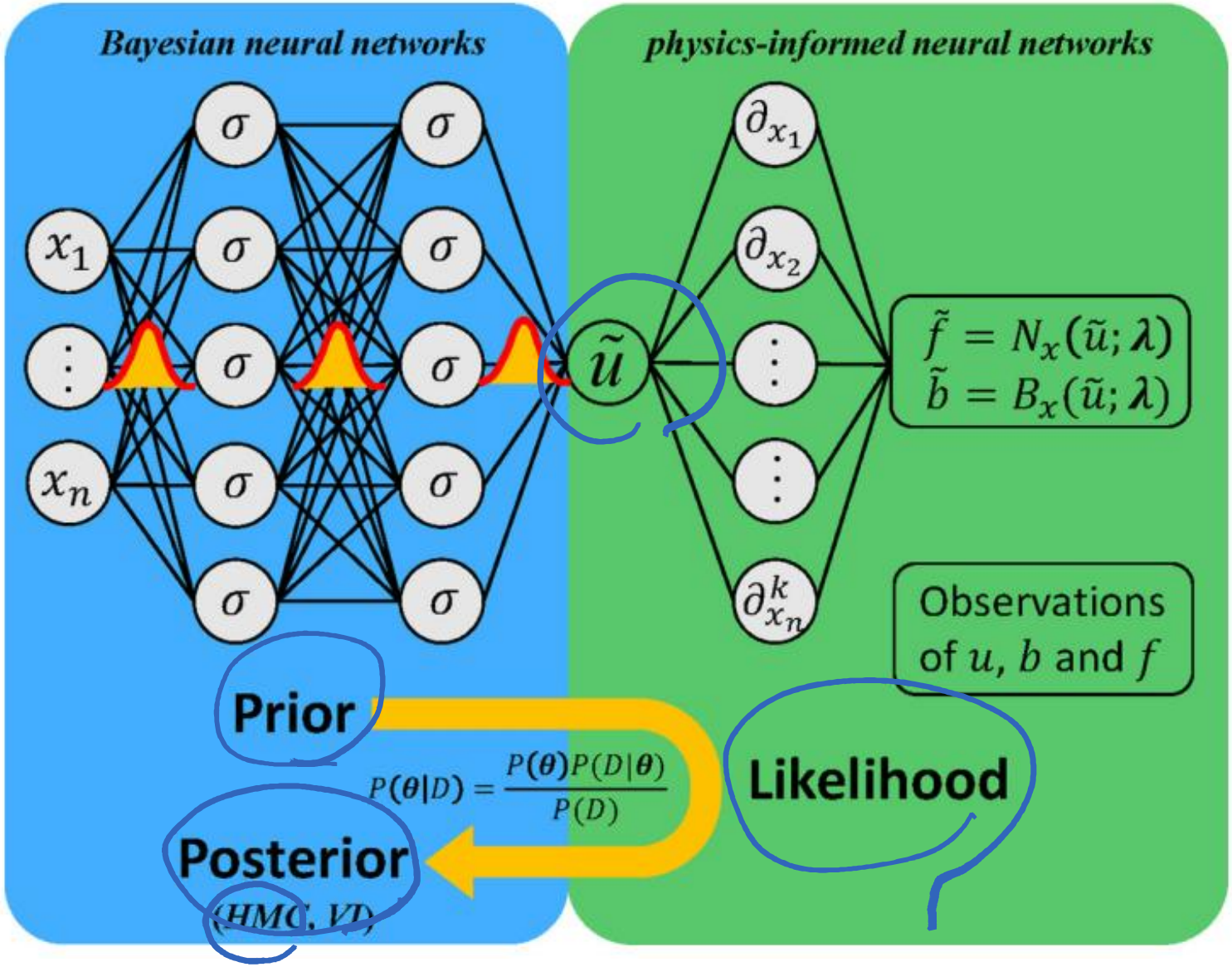
$\theta_2 \rightarrow u_2(x)$

$\theta_3 \rightarrow u_3(x)$

[Prediction]

Prediction : $E[\tilde{u}(x)]$ with **Uncertainty:** $\mathrm{var}(\tilde{u}(x))$

Pipeline 2/2


Bayesian neural networks
physics-informed neural networks
x_1
x_n
σ
$\tilde{u}$
∂_{x_1}
∂_{x_2}
$\partial_{x_n}^k$
$\tilde{f} = N_x(\tilde{u}; \boldsymbol{\lambda})$
$\tilde{b} = B_x(\tilde{u}; \boldsymbol{\lambda})$
Observations of u, b and f
Prior
$P(\boldsymbol{\theta}|D) = \frac{P(\boldsymbol{\theta})P(D|\boldsymbol{\theta})}{P(D)}$
Likelihood
Posterior
(HMC, VI)

Hamiltonian Monte Carlo (HMC) 1/2

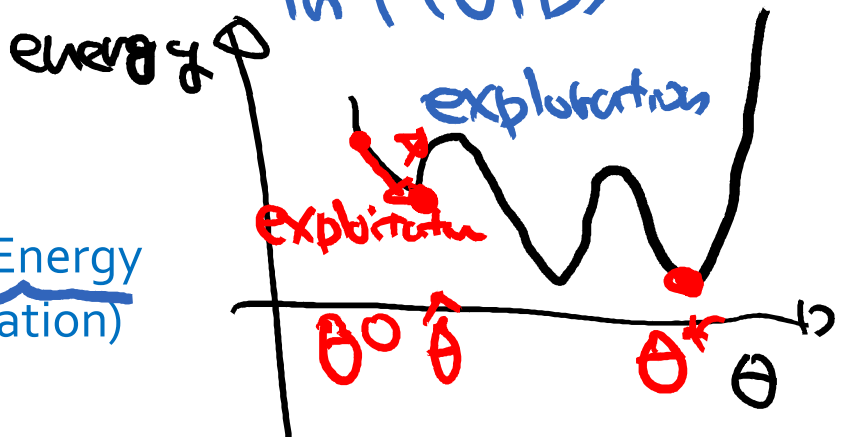
- **Goal of HMC**
  - Sample from the posterior distribution of neural network parameters : $\{\theta^{\{i\}}\} \sim P(\theta \mid D)$
- **Idea of HMC**
  - Probability→ Energy

$$P(\boldsymbol{\theta}|\mathcal{D}) \simeq \exp(-U(\boldsymbol{\theta})), \quad \text{where} \quad U(\theta) = -\ln P(\mathcal{D}|\boldsymbol{\theta}) - \ln P(\boldsymbol{\theta}).$$

  - Auxiliary momentum variable r

$$H(\boldsymbol{\theta}, \boldsymbol{r}) = U(\boldsymbol{\theta}) + \frac{1}{2}\boldsymbol{r}^T \boldsymbol{M}^{-1}\boldsymbol{r},$$

Potential Energy (exploitation) + Kinetic Energy (exploration)



Hamiltonian Monte Carlo (HMC) 2/2

- **Idea of HMC**
  - Hamiltonian dynamics

$$d\boldsymbol{\theta} = \boldsymbol{M}^{-1}\boldsymbol{r}dt,$$ θ changes according to the momentum

$$d\boldsymbol{r} = -\nabla U(\boldsymbol{\theta})dt.$$ r changes according to the gradient of $U(\theta)$

$\nabla U(\theta)$: direction of increasing energy

$-\nabla U(\theta)$: force toward lower energy

(θ', r')

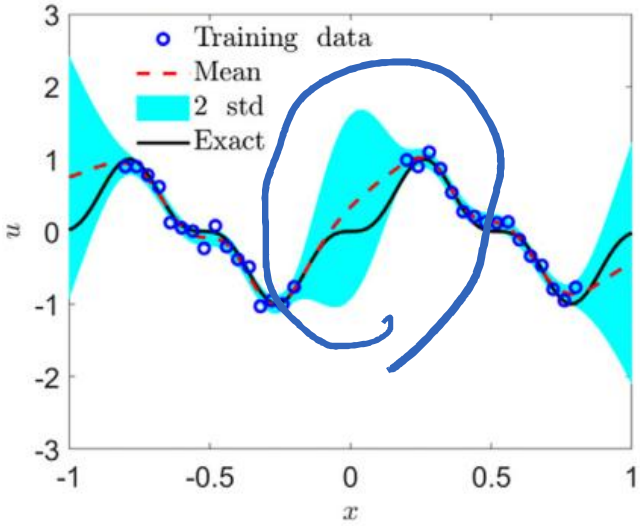
$H(\theta, r) - H(\theta', r')$

$\alpha \leftarrow \min\{1, \exp(H(\boldsymbol{\theta}_L, \boldsymbol{r}_L) - H(\boldsymbol{\theta}^{t_{k-1}}, \boldsymbol{r}^{t_{k-1}}))\}.$
if $p \geq \alpha$ **then**
  $\boldsymbol{\theta}^{t_k} \leftarrow \boldsymbol{\theta}_L,$
else
  $\boldsymbol{\theta}^{t_k} \leftarrow \boldsymbol{\theta}^{t_{k-1}}.$
end if

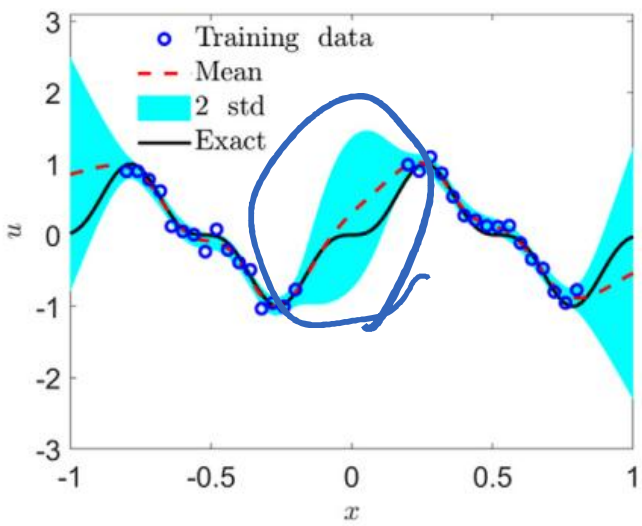
These dynamics approximately preserve the **total energy** (balancing between exploitation and exploration)

Experiments 1/5

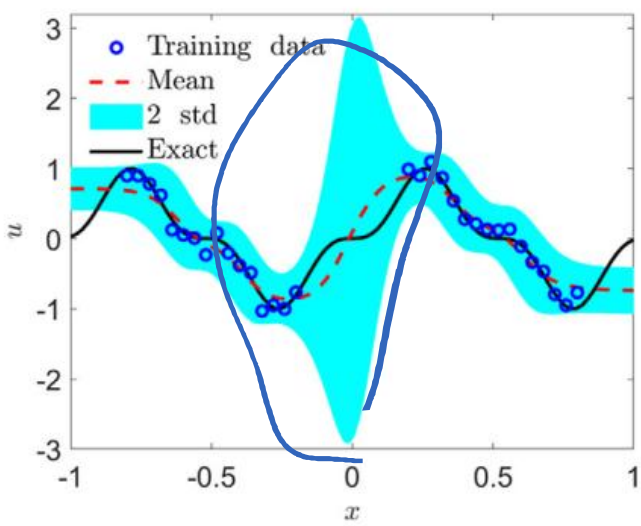
- **Function Regression** $u(x) = \sin^3(6x),\ x \in [-1, 1],$
  - 32 training points with observation noise $\epsilon_u \sim \mathcal{N}(0, 0.1^2)$



BNN-GPR



BNN-HMC



BNN-VI

Experiments 2/5

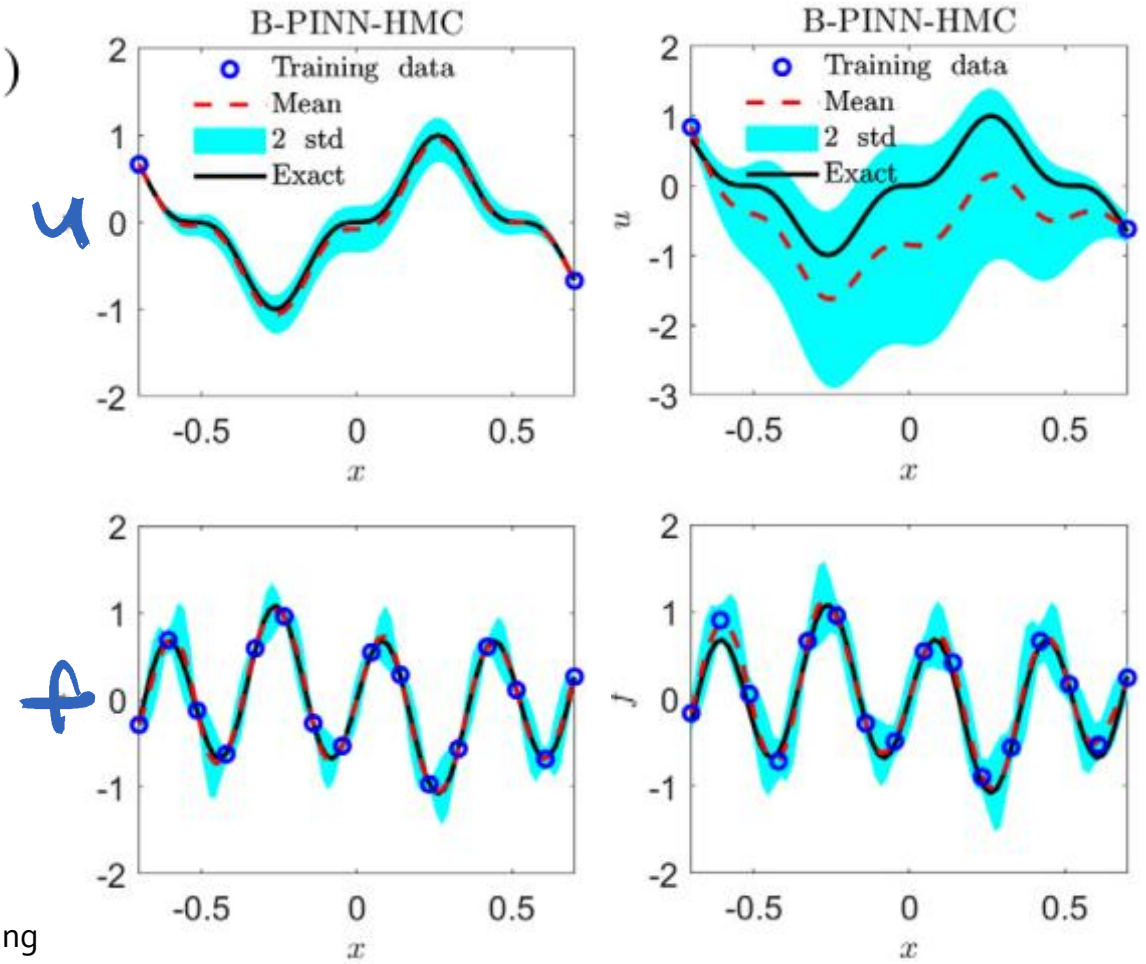
- **1D Poisson Equation** $\lambda \partial_x^2 u = f, x \in [-0.7, 0.7],$
  - 16 sensors for f, 2 sensors for u
  - Noise

$\epsilon_f \sim \mathcal{N}(0, 0.01^2), \epsilon_b \sim \mathcal{N}(0, 0.01^2)$

$\epsilon_f \sim \mathcal{N}(0, 0.1^2), \epsilon_b \sim \mathcal{N}(0, 0.1^2)$

Noise ↓

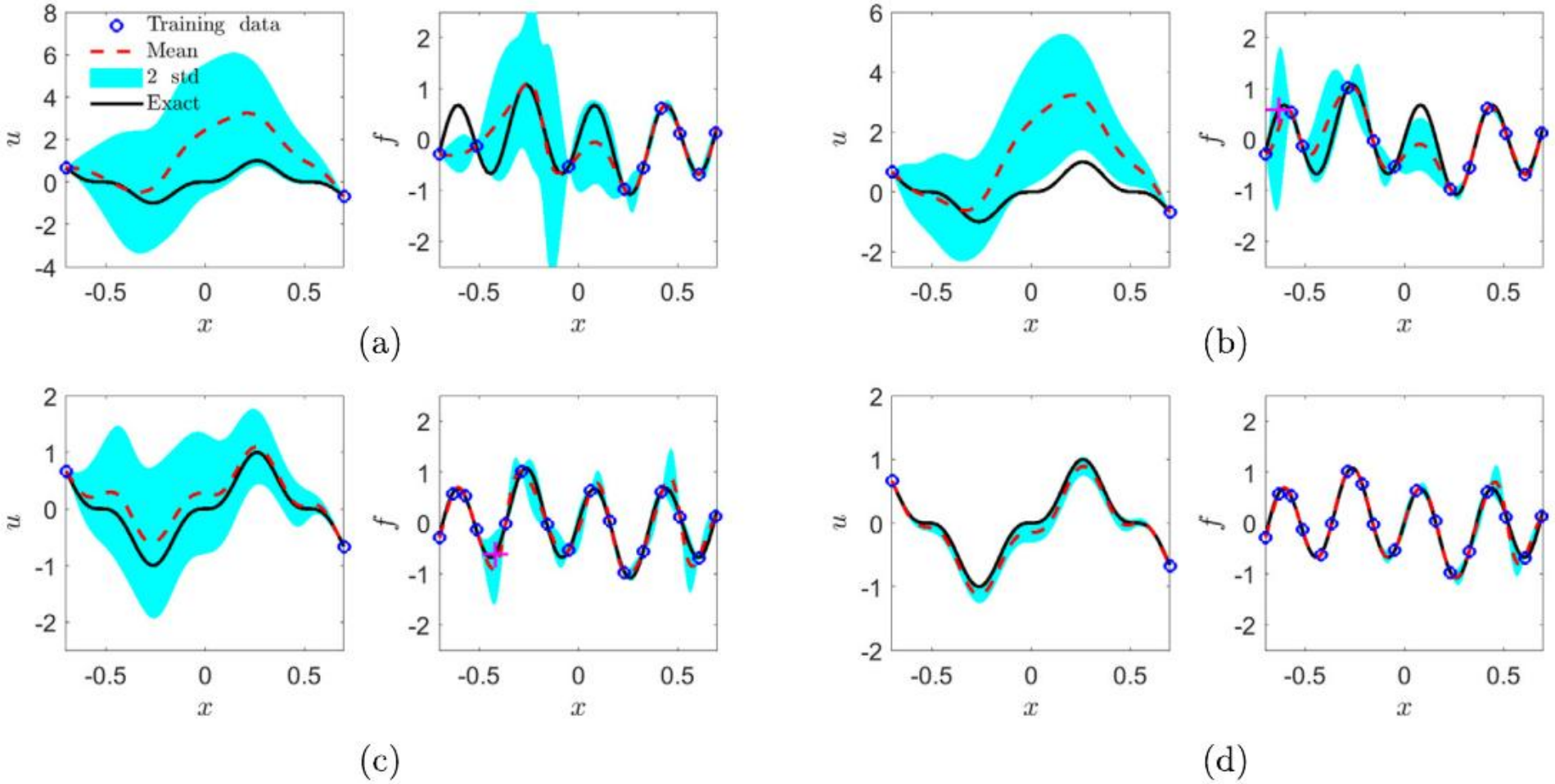
Noise ↑



Experiments 3/5

- **1D Poisson Equation** $\lambda \partial_x^2 u = f, x \in [-0.7, 0.7],$

(a) N = 8, (b) N = 12, (c) N = 16, (d) N = 18



Experiments 4/5

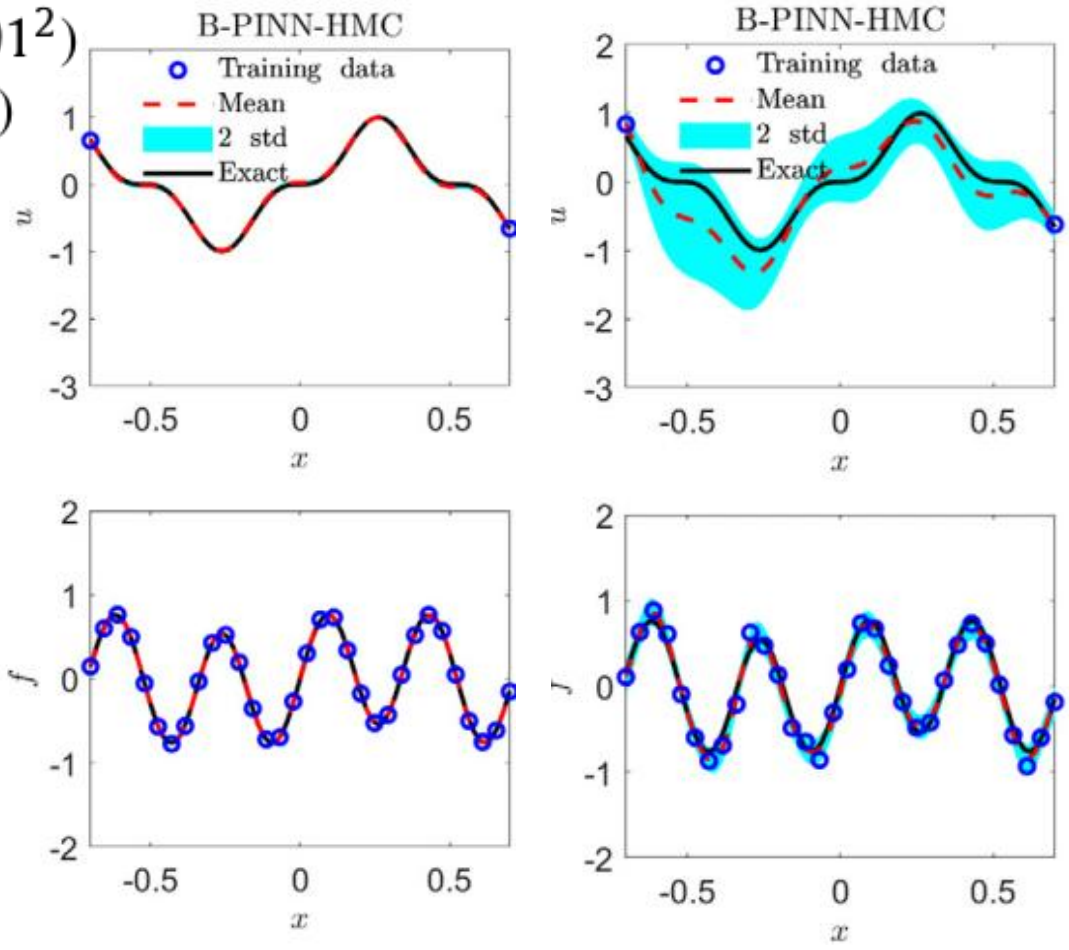
- **1D nonlinear Poisson equation**

$$\lambda \partial_x^2 u + k \tanh(u) = f, \; x \in [-0.7, 0.7],$$

  - 16 sensors for f, 2 sensors for u
  - Noise

$\epsilon_f \sim \mathcal{N}(0, 0.01^2)$, $\epsilon_b \sim \mathcal{N}(0, 0.01^2)$

$\epsilon_f \sim \mathcal{N}(0, 0.1^2)$, $\epsilon_b \sim \mathcal{N}(0, 0.1^2)$



Experiments 5/5

- **2D nonlinear Allen-Cahn equation**

$$\lambda(\partial_x^2 u + \partial_y^2 u) + u(u^2 - 1) = f,\ x, y \in [-1, 1],$$

  - 500 sensors for f, 25 sensors for u
  - Noise

$$\epsilon_f \sim \mathcal{N}(0, 0.01^2),\ \epsilon_b \sim \mathcal{N}(0, 0.01^2)$$
$$\epsilon_f \sim \mathcal{N}(0, 0.1^2),\ \epsilon_b \sim \mathcal{N}(0, 0.1^2)$$

